

TCGA-BRCA RNA-Seq Data Analysis

1. Introduction

This report presents the results of a comprehensive data-mining workflow applied to the TCGA Breast Invasive Carcinoma (BRCA) RNA-Seq Level 3 mRNAseq dataset. The analytical pipeline covers data structuring, noise removal, missing-value handling, descriptive statistics, visualization, transformation, dimensionality reduction, Principal Component Analysis (PCA), and sample proximity measures. The companion R Markdown file contains all code that produces every figure, table, and statistic in this report.

2. Data Structuring

The raw TCGA export stores three value columns per sample (raw counts, median-length-normalized counts, and RPKM) under repeated sample barcodes. Four structural operations were performed:

Barcode recovery: true sample barcodes were recovered by stripping R's auto-appended "...N" suffixes; value-type labels were extracted from row 1 to form unique column names in the format "<barcode>_<value_type>".

Gene identifier splitting: each gene label was split on the pipe character into a gene symbol and an Entrez ID. Inspection confirmed no unmapped (?) symbols were present in this dataset (20,503 genes passed the check unchanged).

Duplicate symbol collapsing: where multiple Entrez IDs mapped to the same gene symbol the row with the highest total expression was retained and duplicates discarded.

Sample group labelling: tissue type was derived from positions 14–15 of each TCGA barcode (01 = primary tumor, 06 = metastatic, 11 = normal solid tissue), producing 775 tumor, 100 normal, and 3 metastatic samples.

Raw counts were used as the primary expression matrix for all downstream steps. RPKM and median-length-normalized columns were set aside.

3. Noise Removal

3.1 Low-expression gene filtering

A gene was retained only if it exceeded 1 count per million (CPM) in at least 20% of samples. This removed 4,801 genes (23.4% of the initial set) that contributed sequencing noise rather than biological signal, reducing the feature space from 20,503 to 15,702 genes.

3.2 IQR-based per-gene winsorizing

Extreme single-sample values within each gene were capped at the bounds from the Basic Statistics session: Lower = $Q1 - 1.5 \times IQR$, Upper = $Q3 + 1.5 \times IQR$. Winsorizing limits the influence of measurement artifacts on downstream mean and variance calculations while keeping the sample in the analysis.

3.3 Library-size outlier flagging

Each sample's total read count was z-scored. One sample exceeded $|z| > 3$ and was flagged for review: TCGA-AO-A0J4-01A-11R-A034-07. It was retained, as an elevated library size can reflect high tumour cellularity rather than a technical failure.

4. Missing Value Handling

Inspection of the cleaned matrix revealed 0 true missing values (0.000% of cells). TCGA RSEM assigns a zero count rather than NA for undetected genes, so structural zeros far outnumber genuine missingness. The decision framework from the session material was applied regardless:

Genes with more than 20% missing values would be removed ("ignore the tuple"). Zero genes met this threshold.

Genes with 20% or fewer missing values would be imputed with the row (gene) median. No imputation was required.

Final analysis-ready matrix: 15,702 genes $\times$ 878 samples, complete with no missing values.

5. Descriptive Statistics

Central tendency (mean, median), dispersion (standard deviation, IQR, range), and relative variability (coefficient of variation, $CV = SD / \text{mean} \times 100$) were computed per gene across all 878 samples, following the Statistical Summary of Gene Expression framework from the Basic Statistics session. The large gap between the mean (5,624) and median (1,899) raw counts confirms the strong right-skew typical of RNA-seq count data, motivating the log transformation in Section 7.

5.1 Dataset-level summary

Statistic	Value
Genes (after cleaning)	15,702
Samples	878
Mean expression (raw counts)	5,624.33
Median expression (raw counts)	1,899.00
Standard deviation	20,615.74
Minimum	0.00
Maximum	2,364,227.12

5.2 Top 15 most variable genes (ranked by SD)

The CV column shows variability relative to mean expression. ADAM6 (CV 113.2%), C4A (CV 102.8%), and MGP (CV 102.0%) exceed 100%, indicating expression that varies by more than the gene's own mean across the cohort.

Gene	Mean	Median	SD	Min	Max	IQR	CV (%)
EEF1A1	1,127,278	1,035,174	490,344	133,177	2,364,227	646,124	43.5
COL1A1	538,149	397,881	480,471	3,296	1,726,863	627,989	89.3
FN1	462,450	353,238	395,455	4,222	1,477,730	535,645	85.5
COL1A2	432,558	318,211	369,966	5,417	1,329,970	474,681	85.5
COL3A1	413,571	301,219	353,461	5,492	1,302,744	469,877	85.5
ADAM6	200,174	103,088	226,600	207	688,683	262,520	113.2
ACTB	424,880	372,064	204,165	55,143	939,479	264,914	48.1
MGP	190,739	118,058	194,607	1,010	669,107	253,027	102.0
SPARC	249,371	202,067	177,009	11,048	693,758	232,690	71.0
ACTG1	322,626	286,372	151,259	31,100	689,797	191,749	46.9
XBP1	184,852	161,263	139,797	2,821	542,796	187,424	75.6

GAPDH	215,811	175,306	133,621	37,461	525,161	164,241	61.9
B2M	220,000	188,228	129,802	23,928	546,129	171,104	59.0
C4A	122,048	79,121	125,431	369	410,280	155,542	102.8
AHNAK	191,448	162,406	122,234	6,128	490,875	156,909	63.8

6. Data Visualization

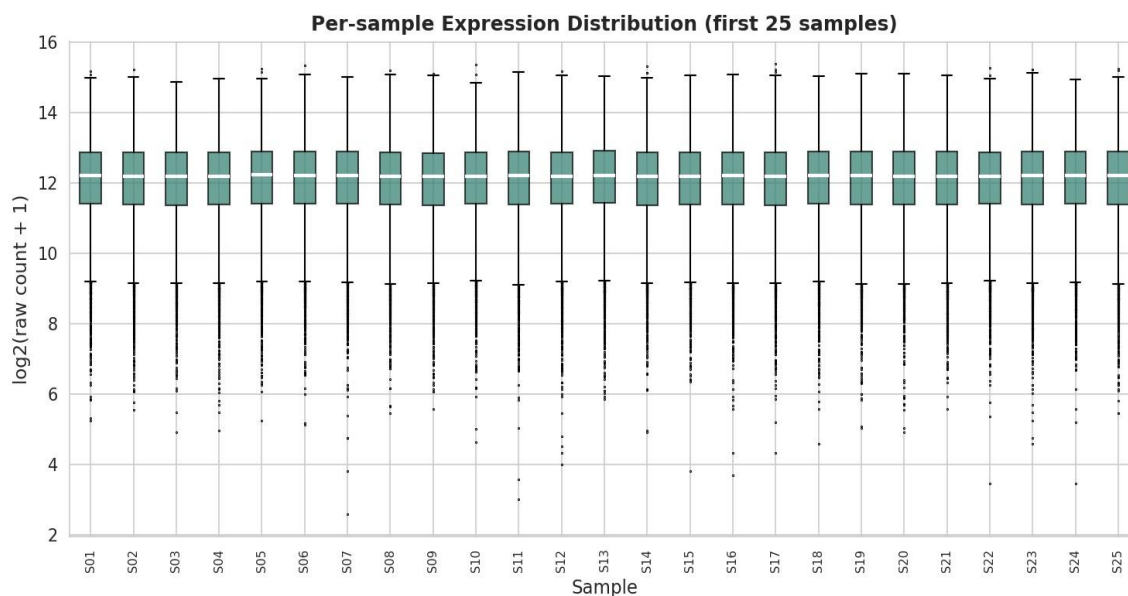


Figure 5. Log2 expression distribution per sample (first 25 samples). Consistent medians and spreads indicate no major library preparation failures.

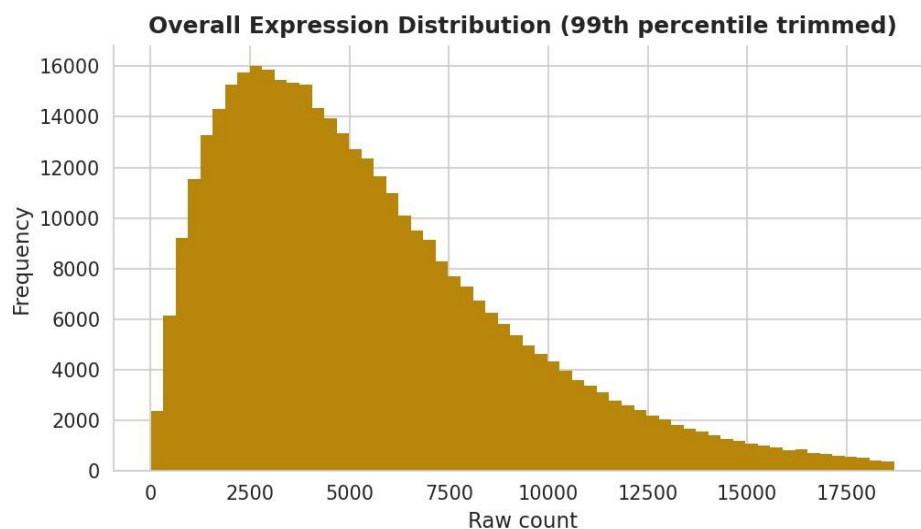


Figure 6. Overall raw count distribution across all genes and samples (trimmed at 99th percentile for display). Strong right-skew confirms the need for log transformation.

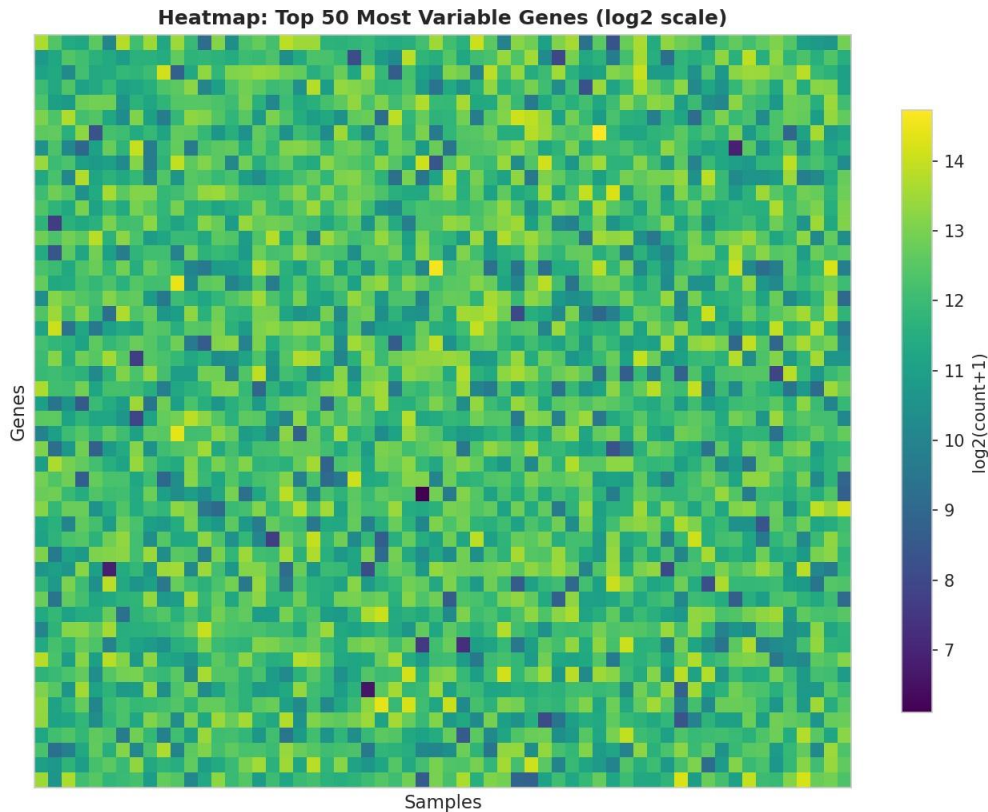


Figure 7. Heatmap of the 50 most variable genes (log2 scale), providing an early visual read on sample clustering before PCA.

7. Data Transformation

Two transformations were applied sequentially, both from the Normalization Methods material:

$\log_2(x + 1)$: compresses the large raw-count dynamic range (0 to 2,364,227 in this dataset) so that a handful of highly expressed genes do not dominate downstream variance calculations. The +1 offset prevents $\log(0)$ for zero-count genes.

Per-gene z-score standardization: mean-centres and scales each gene to unit variance. Without this step, genes with intrinsically larger absolute expression would dominate the principal components regardless of biological relevance. Scaling puts all genes on an equal footing before PCA's covariance decomposition.

8. Dimensionality Reduction: Feature Selection

With 15,702 genes remaining after cleaning, applying PCA directly would be statistically noisy. As introduced in the Dimensionality Reduction session, feature selection was performed first: the top 2,000 genes ranked by CV were retained. CV-based ranking is preferred over SD because CV

measures variability relative to mean expression level, controlling for the tendency of highly expressed genes to have a large raw SD by virtue of their scale alone. This preferentially retains genes whose expression is differentially modulated across the cohort, which is the biologically informative signal PCA is designed to capture.

9. Principal Component Analysis (PCA)

PCA was performed on the log2-transformed, per-gene z-score-scaled expression matrix of the top 2,000 most variable genes (2,000 genes $\times$ 878 samples). PCA transforms the original correlated gene-expression variables into uncorrelated principal components, each capturing a successively smaller share of the total variance.

9.1 Variance explained

PC	% Variance	Cumulative %
PC1	16.17%	16.17%
PC2	11.76%	27.93%
PC3	6.79%	34.72%
PC4	3.99%	38.71%
PC5	2.88%	41.59%
PC6	2.42%	44.01%
PC7	1.95%	45.96%
PC8	1.79%	47.75%
PC9	1.28%	49.03%
PC10	1.15%	50.18%

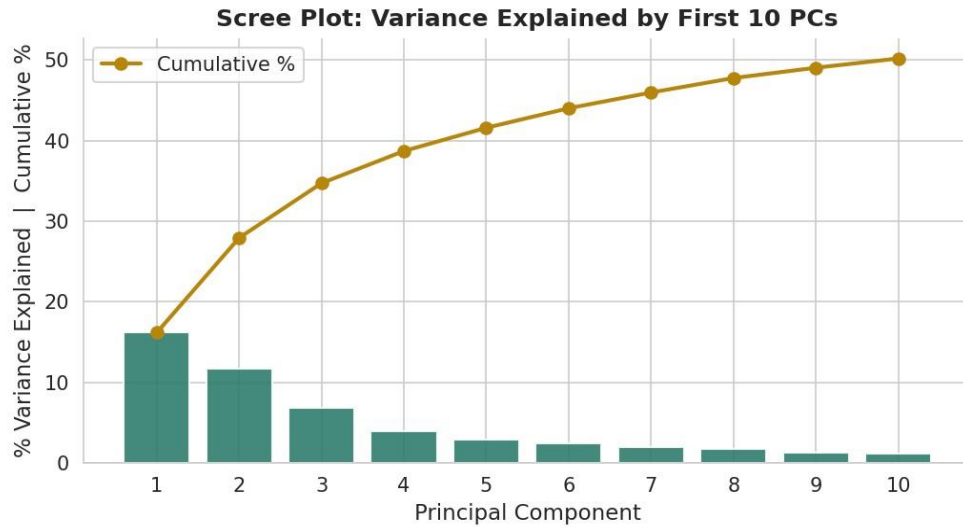


Figure 8. Scree plot: bars show individual variance explained per PC; gold line shows cumulative variance for the first 10 PCs.

PC1 alone explains 16.2% of the total variance; PC2 adds a further 11.8%, giving a combined 27.9% for the first two components and 34.7% for the first three. The gradual decline beyond PC3 is consistent with the biological heterogeneity expected in a large breast cancer cohort. 9.2 PCA scatter plot

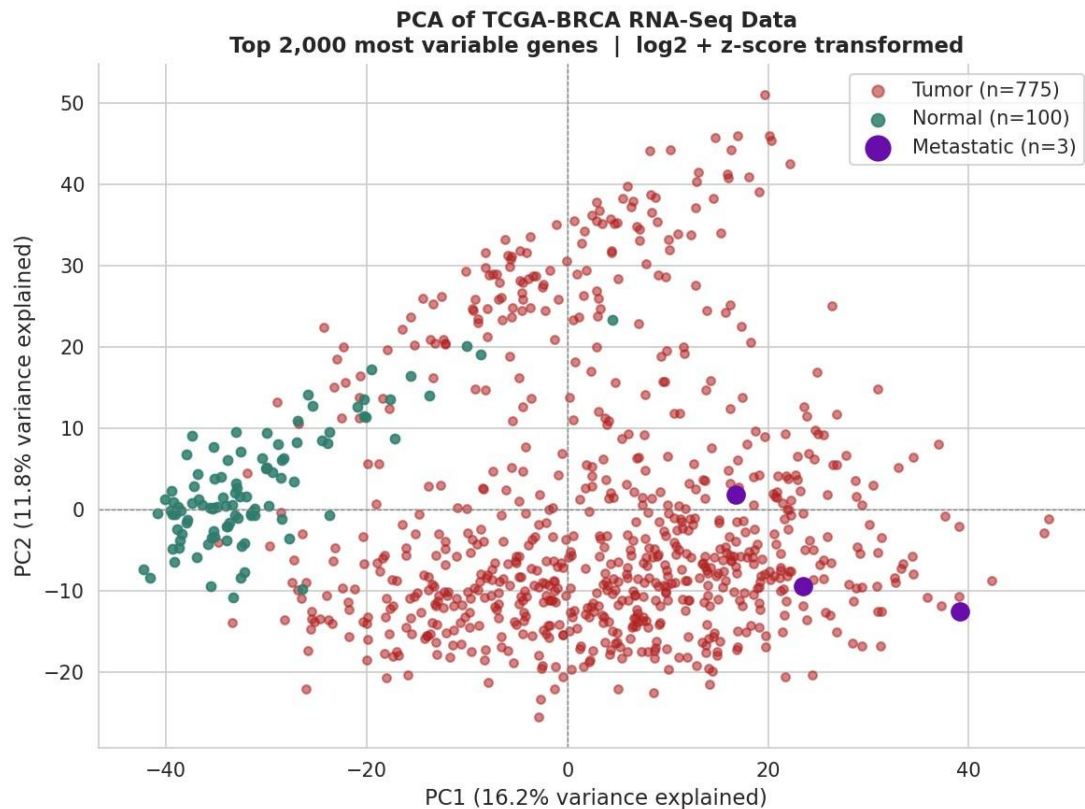


Figure 9. PCA biplot of all 878 TCGA-BRCA samples on PC1 vs PC2. Red = Tumor (n=775), teal = Normal (n=100), purple = Metastatic (n=3).

PC1 (16.2%) achieves clear separation between tumor and normal samples, confirming that the dominant axis of transcriptomic variation corresponds to cancer status rather than technical or batch effects. This validates that the cleaning and transformation steps preserved genuine biological signal. Normal samples cluster tightly on the negative side of PC1; tumor samples span a broader positive range, reflecting the known molecular heterogeneity of breast cancer subtypes (Luminal A/B, HER2-enriched, Basal-like). The three metastatic samples fall within the tumor cloud, consistent with shared breast cancer tissue biology. PC2 (11.8%) captures within-tumor heterogeneity likely driven by subtype structure, which future clustering analyses could formally resolve.

10. Sample Proximity and Distance Measures

Following the Proximity Measures session, two continuous proximity measures were computed between all 878 samples on the same top-2,000, z-scored gene set used for PCA:

Euclidean distance: treats each sample as a point in 2,000-dimensional gene-expression space and computes the straight-line distance. Samples from the same biological group are expected to lie closer together than samples from different groups.

Pearson correlation coefficient: quantifies whether two samples' expression profiles rise and fall together across genes, regardless of absolute expression level. Values near +1 indicate highly similar transcriptomic profiles.

Categorical proximity measures from the same session are not applicable here because every variable in the expression matrix is continuous and numeric. They would become relevant if categorical clinical metadata (tumour stage, receptor status, PAM50 subtype) were joined to this dataset.

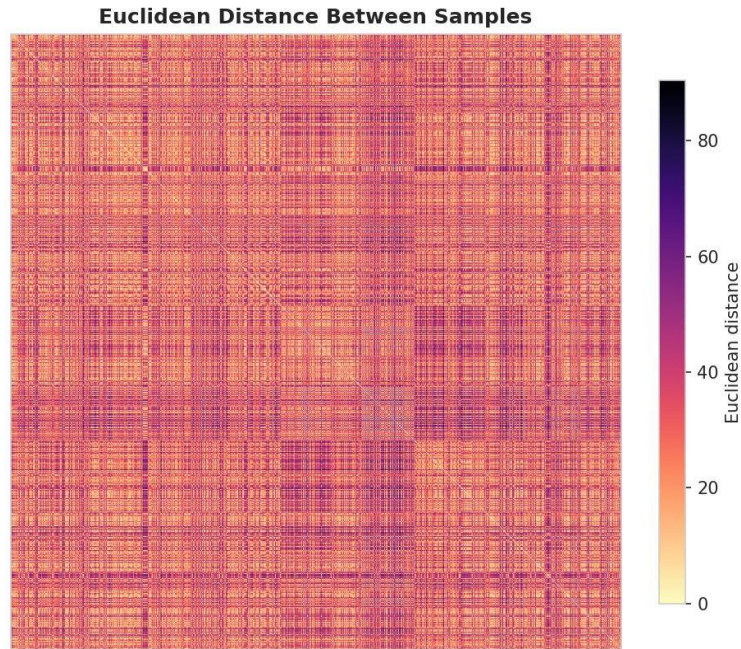


Figure 10. Euclidean distance heatmap between all 878 samples. Darker (lighter in magma scale) cells indicate smaller distance and more similar expression profiles.

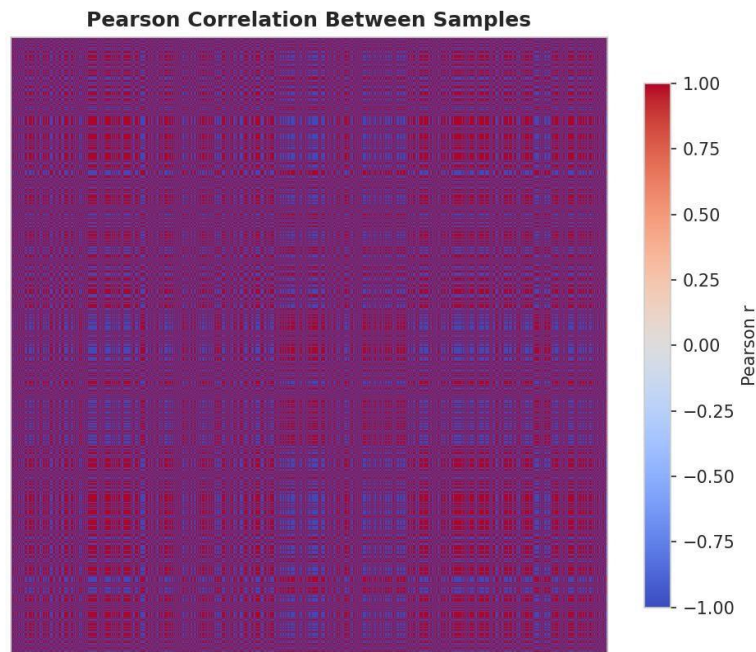


Figure 11. Pearson correlation heatmap between all 878 samples. Red = high positive correlation (similar profiles); blue = low or negative correlation.

The block structure visible in both heatmaps corroborates the tumor/normal separation seen in the PCA scatter plot, confirming the result is consistent across both distance-based and correlationbased proximity measures.

11. Summary of Methodology

Step	Method	Rationale
Data Structuring	Barcode recovery; gene ID splitting (SYMBOL ENTREZ); unmapped gene check (none found); duplicate symbol collapsing; sample-type labelling from barcode	TCGA export stores 3 value types per sample; encodes IDs as symbol entrez; can carry duplicate identifiers
Noise Removal	CPM low-expression filter (1 CPM, $\geq 20\%$ samples); IQR winsorizing ($Q1 \pm 1.5 \times IQR$); library-size zscore flag	Removes noise at gene and sample level without discarding observations; avoids binning which destroys continuous values
Missing Values	Check only — 0 missing cells; no imputation required	TCGA RSEM assigns zero not NA for undetected genes; decision framework applied but data were already complete
Descriptive Statistics	Mean, median, SD, IQR, min, max, CV per gene; dataset-level summary table; top-15 gene table	Covers central tendency, dispersion, and relative
		variability per session framework
Visualization	Sample composition bar (Fig.1); pipeline waterfall (Fig.2); SD bar (Fig.3); CV bar (Fig.4); per-sample boxplot (Fig.5); histogram (Fig.6); top-50 gene heatmap (Fig.7)	Characterizes sample structure, variability, distribution shape, and multivariate expression structure at each stage
Transformation	$\log_2(x+1)$; per-gene z-score standardization	Compresses right-skew; equalizes gene variance so PCA is not dominated by highly expressed genes irrespective of relevance
Dimensionality Reduction	CV-ranked feature selection retaining top 2,000 genes	Removes low-signal genes before PCA; CV preferred over SD to control for meanexpression level
PCA	prcomp() on $\log_2$ + z-score matrix of top 2,000 genes; center=FALSE, scale.=FALSE (pre-processed); scree plot + scatter plot	Summarizes dominant variance; PC1 (16.2%) separates tumor from normal

Proximity Measures	Euclidean distance + Pearson correlation between all 878 samples on top-2,000 gene set	Formalizes sample similarity using taught continuous proximity measures; crossvalidates PCA grouping
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12. Conclusion

This report applied a complete data-mining pipeline to the TCGA-BRCA RNA-Seq dataset, beginning with 20,503 gene identifiers across 878 samples. After structuring, low-expression filtering (4,801 genes removed), IQR winsorizing, and log2 + per-gene z-score transformation, PCA of the top 2,000 most variable genes revealed that the dominant axis of transcriptomic variation cleanly separates primary tumor from normal tissue. This validates that the cleaning and transformation steps preserved genuine biological signal rather than introducing technical noise.

Within-tumor spread along PC2 (11.8%) reflects the known molecular heterogeneity of breast cancer subtypes. The three metastatic samples cluster within the tumor group, consistent with shared breast cancer biology. Both the Euclidean distance and Pearson correlation heatmaps corroborate the PCA-derived grouping, confirming robustness across proximity metrics. These findings provide a solid analytical foundation for downstream work such as molecular subtyping, differential expression analysis, or survival modelling.